Fitoterapia 111 (2016) 58–65

Contents lists available at ScienceDirect

Fitoterapia

journal homepage: www.elsevier.com/locate/fitote

New isoindolinones from the fruiting bodies of *Hericium erinaceum*

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ARTICLE INFO

Article history:
Received 22 February 2016
Received in revised form 12 April 2016
Accepted 13 April 2016
Available online 16 April 2016

Keywords:
Hericium erinaceus
Fruiting bodies
Isoindolinone
Erinaceolactams A-E
Cytotoxicity

ABSTRACT

Hericium erinaceus is a well-known medicinal and edible mushroom, which is considered as a potential source to obtain antitumor candidates. In this work, five new isoindolinones, named erinaceolactams A-E (**1**–**5**), along with five known compounds (**6**–**10**), were isolated from 70% ethanol extract of the fruiting bodies of *H. erinaceus*. The structures of new compounds were validated by HRESIMS and 1D, 2D NMR. It's worth mentioning that there are two pairs of isomers included in the new compounds. Moreover, their cytotoxicity against metastatic human hepatocellular carcinoma cell lines SMMC-7221 and MHCC-97H were evaluated. The results showed that compounds **6** and **7** exhibited promising inhibitory potency against the growth of two cell lines.
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1. Introduction

Hericium erinaceus (also called "Houtougu" in Chinese for its similar appearance to monkey's head) is well known in Asian countries, such as China, Japan, Korea and so on. As a basidiomycetous fungus, the mushroom belongs to the family Hericiaceae in the order Russulales and grows on dead trunks of hard wood [1]. It impresses the public because of its good taste as a dish and its application as an herbal medicine [2,3]. In traditional Chinese medicine, *H. erinaceus* is used to treat enervation and gastrointestinal diseases as dyspepsia, gastric ulcers [4]. For the successful cultivation of *H. erinaceus*, it is now widely accessible as a functional food to possess various pharmacological effects including antmicrobial, anti-tumor, immunomodulatory and neuroprotection [5]. Over the past decades, the chemical constituents of *H. erinaceus* have been explored, which range from macromolecules such as glycoprotein [6] and polysaccharides [7–9] to small molecules including sterols [10], fatty acids [11], hericerins [12], erinacines [13], erinacerins [14] and erinaceolactones [15]. In our previous work, some extracts of *H. erinaceus* were revealed to display inhibition activity against gastric cancer NCI-87 cells in vitro and tumor xenografts bearing in SCID mice in vivo, and these extracts had the potential for development into anticancer agents for the treatment of gastrointestinal cancer used alone and/or in combination with 5-fluorouraci [16]. In order to compare fruiting body with mycelium and screen out the active compounds, we continued to perform the isolation of the extract from the fruiting bodies of *H. erinaceus* in present study, and also investigated the cytotoxicity activity against human liver cancer cell lines SMMC-7221 and MHCC-97H in vitro.

2. Materials and methods

2.1. General experimental procedures

Optical rotations were obtained on a Jasco model 1020 polarimeter (Horiba, Tokyo, Japan). High resolution electrospray ionization mass spectroscopy (HRESIMS) and collision induced dissociation (CID) experiments were performed in an Agilent Technologies liquid chromatograph connected to Q-TOF mass spectra. NMR spectra were recorded on Avance III 400 MHz and Avance III 500 MHz Bruker spectrometers (Bruker BioSpin, Rheinstetten, Germany). DMSO-d_6 was used as solvent, and the chemical shifts are given in parts per million referring to the residual nondeuterated solvent signal (DMSO-d_6 2.50 and 39.52 ppm for ^{1}H and ^{13}C, respectively). Silica gel (80–100 and 200–300 mesh) for column chromatography (CC) and silica gel GF-254 for thin layer chromatography (TLC) were purchased from the Qingdao Marine Chemical Factory (Qingdao, China). Pre-HPLC experiments were conducted using an Agilent 1200 HPLC system and carried on preparative YMC Pack ODS-A (10 μm, 250 × 10 mmL, YMC Co., Ltd, Kyoto, Japan).

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http://dx.doi.org/10.1016/j.fitote.2016.04.010
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2.2. Fungal material

Dried fruiting bodies of *H. erinaceus* were purchased from a mushroom market of Gutian, Fujian province, China. It was identified by one of the authors (Kangping Xu). A voucher specimen was deposited at the Herbarium library of the School of Pharmaceutical Sciences, Central South University.

2.3. Extraction and isolation

Dried fruiting bodies (5 kg) of *H. erinaceus* were smashed and extracted twice with 70% ethanol (20 L) under reflux. The 70% ethanol extract was concentrated under reduced pressure, and the residue (400 g) was partitioned into H_2O and extracted with pertroleum ether and EtOAc successively, yielding pertroleum ether (42 g) and EtOAc (54 g) fractions. The EtOAc fraction (54 g) was subjected to silica gel column chromatography with a gradient of pertroleum ether-EtOAc-MeOH (50:1:0, 20:1:0, 10:1:0, 5:1:0, 2:1:0, 2:1:0.5, 2:1:1, MeOH) to give eight fractions (1A–1H). Fraction 1C was subjected to YMC (250 × 10 mmL) column chromatography with ACN–H_2O to yield compound **7** (2 mg). Fraction 1D was subjected to YMC (250 × 10 mmL) column chromatography with ACN–H_2O to yield compound **9** (17 mg) and **10** (24 mg). Fraction 1E (3.1 g) was chromatographed over silica gel repeatedly and then recrystallized to yield compound **6** (30 mg) and **8** (20 mg). Fraction 1F (5.4 g) was chromatographed over silica gel with a gradient of pertroleum ether-EtOAc-MeOH (2:1:0, 2:1:0.1, 2:1:0.3, 1:1:0.3, 1:1:0.5, 1:1:1, MeOH) to give seven fractions (2A–2G). Fraction 2C was then recrystallized to yield compound **1** (14 mg). Fraction 2D was subjected to YMC (250 × 10 mmL) column chromatography with ACN–H_2O to yield compound **3** (4 mg). Fraction 2E was subjected to YMC (250 × 10 mmL) column chromatography with ACN–H_2O to yield compounds **2** (15 mg) and **4** (12 mg). Fraction 2E was subjected to YMC (250 × 10 mmL) column chromatography with ACN–H_2O to yield compound **5** (7 mg).

Erinaceolactam A (**1**): colorless cluster crystal; HRESIMS m/z 330.1703 $[M + H]^+$ (calcd for $C_{19}H_{24}NO_4$, 330.1705). For ^1H NMR (500 MHz, DMSO-d_6) and ^{13}C NMR (125 MHz, DMSO-d_6) spectroscopic data, see Table 1.

Erinaceolactam B (**2**): colorless oil; HRESIMS m/z 373.1887 $[M]^+$ (calcd for $C_{21}H_{27}NO_5$, 373.1889). For ^1H NMR (500 MHz, DMSO-d_6) and ^{13}C NMR (125 MHz, DMSO-d_6) spectroscopic data, see Table 1.

Erinaceolactam C (**3**): white amorphous powder; $[\alpha]_D^{20} = 0$ ($c = 0.1$, MeOH); HRESIMS m/z 330.1704 $[M + H]^+$ (calcd for $C_{19}H_{24}NO_4$, 330.1705). For ^1H NMR (500 MHz, DMSO-d_6) and ^{13}C NMR (125 MHz, DMSO-d_6) spectroscopic data, see Table 1.

Erinaceolactam D (**4**): yellow amorphous powder; $[\alpha]_D^{20} = 0$ ($c = 0.1$, MeOH); HRESIMS m/z 374.1965 $[M + H]^+$ (calcd for $C_{21}H_{28}NO_5$, 374.1967). For ^1H NMR (500 MHz, DMSO-d_6) and ^{13}C NMR (125 MHz, DMSO-d_6) spectroscopic data, see Table 1.

Erinaceolactam E (**5**): yellow amorphous powder; $[\alpha]_D^{20} = 0$ ($c = 0.1$, MeOH); HRESIMS m/z 415.2245 $[M + H]^+$ (calcd for $C_{23}H_{31}N_2O_5$, 415.2233). For ^1H NMR (500 MHz, DMSO-d_6) and ^{13}C NMR (125 MHz, DMSO-d_6) spectroscopic data, see Table 1.

2.4. Cell growth inhibition assay

Human liver cancer cell line SMMC-7221 and MHCC-97H were purchased from American Type of Culture Collection (ATCC). The cells were cultured at 37 °C under 5% CO_2 in Dulbecco's modified Eagle's medium (DMEM) containing 10% fetal bovine serum (FBS). Cell viability analysis was performed by MTT assay. SMMC-7221 and MHCC-97H cells were cultured in 96-well plates at cell density 8000 cells/well. After one-day cultivation, the cells were cultured in DMEM without FBS and varying concentrations (5 μg/mL, 10 μg/mL, 20 μg/mL) of samples **1**–**12**. 5-Fluoro-2.4-(1 h, 3 h)-pyrimidinedione (5-FU, 5 μg/mL) was used as positive control which was dissolved in dimethyl sulfoxide (DMSO). Controls were done with the same final DMSO concentration in the medium as samples. After incubation of the cells with samples 1–12

Table 1
^1H and ^{13}C NMR data of compounds **1**–**5** in DMSO-d_6.

Position	**1**		**2**		**3**		**4**		**5**	
	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)
1	170.7		168.1		170.6		168.0		168.1	
2		8.39(1H, s)				8.40(s)				
3	43.5	4.20(2H, s)	49.3	4.39 (2H, s)	42.6	4.06(d,16.8)	48.4	4.26(d, 17.5)	47.3	4.16(d, 17.4)
						4.13(d, 16.8)		4.32(d, 17.5)		4.24(d, 17.4)
3a	123.4		121.6		132.5		132.4		132.4	
4	150.6		150.3		149.0		148.7		148.8	
5	119.7		119.7		113.5		113.3		113.4	
6	158.8		158.8		158.4		158.4		158.5	
7	96.7	6.75(1H, s)	96.8	6.76 (1H, s)	96.1	6.76(s)	96.1	6.78(s)	96.2	6.77(s)
7a	132.1		132.1		124.5		122.5		122.2	
1′	23.0	3.35(2H, s)	23.0	3.36 (2H, s)	17.4	2.62(t, 6.7)	17.4	2.62(t, 6.7)	17.4	2.62(t, 6.7)
2′	127.0	5.24(1H, dd, 6.0, 6.5)	127.1	5.23 (1H, dd, 6.5, 7.5)	30.7	1.85(m)	30.7	1.84(m)	30.6	1.84(m)
						1.94(m)		1.93(m)		1.94(m)
3′	129.8		129.8		76.2		76.2		76.2	
4′	55.0	2.99(2H, s)	56.3	2.99 (2H, s)	52.0	2.60(d, 13.8)	52.1	2.60(d, 14.0)	52.0	2.60(d, 14.0)
						2.79(d, 13.8)		2.80(d, 14.0)		2.80(d, 14.0)
5′	199.0		198.9		198.2		198.2		198.2	
6′	123.6	6.10(1H, brs)	123.4	6.09 (H, t, 1.5)	125.6	6.12(m)	125.6	6.12(m)	125.6	6.12(m)
7′	154.9		154.9		154.7		154.9		154.8	
8′	27.6	1.81(3H, s)	27.6	1.81 (3H, d, 1.0)	27.6	1.82(d, 0.85)	27.6	1.83(d, 0.95)	27.6	1.83(s)
9′	16.9	1.71 (3H, s)	16.9	1.70 (3H, s)	24.9	1.33(s)	24.8	1.33(s)	24.9	1.34(s)
10′	20.6	2.03 (3H, s)	20.6	2.03 (3H, d, 0.5)	20.7	2.06(d, 0.80)	20.8	2.06(d, 0.90)	20.8	2.07(s)
1″			45.2	3.55 (2H, d, 5.0)			45.1	3.54(t, 5.2)	41.9	3.48(t, 7.1)
2″			59.9	3.60 (2H, d, 5.0)			59.9	3.59(t, 5.2)	24.4	1.80(m)
3″									32.8	2.05(t, 7.8)
4″									174.1	
4-OH		9.48(1H, s)		3.80 (3H, s)						
6-OMe	56.3	3.80(3H, s)	55.0	9.53 (1H, br)	56.2	3.84(s)	56.2	3.84(s)	56.3	3.84(s)
2″-OH				4.85 (1H, t, 5.0)				4.83(t, 5.2)		
4″-NH_2										6.74(brs)
										7.28(brs)

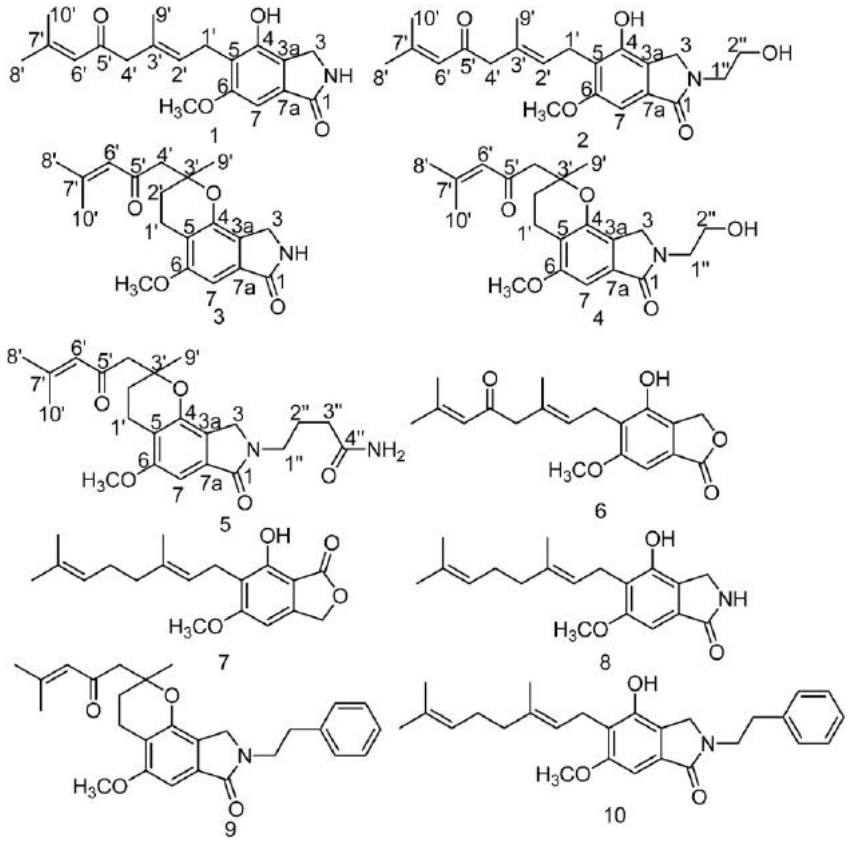


Fig. 1. Structures of compounds **1**–**10** from *Hericium erinaceus.*

for 24 h, 20 μL of MTT (0.5 mg/mL) solution were added to each well and incubated for 4 h at 37 °C. Then, the broth was removed and DMSO (150 μL) was added to each well. The degree of formation, an indicator of living and metabolically active cells, was determined using ELISA reader at 570 nm. The data were expressed as means ± SEM (standard errors). Statistical analysis was performed by unpaired Student's *t*-test for two comparisons. $P < 0.05$ was considered statistically significant.

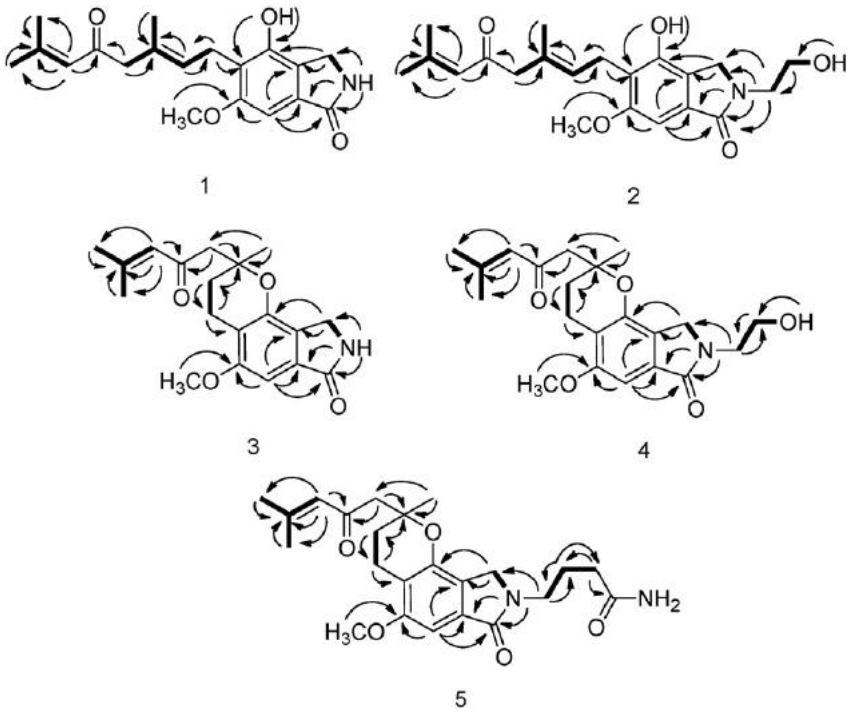


Fig. 2. $^{1}H—^{1}H$ COSY (bold line) and HMBC (H → C) correlations of compounds **1**–**5.**

3. Results and discussion

3.1. Chemistry

Erinaceolactam A (**1**) was obtained as a colorless cluster crystal after crystallization from petroleum ether-EtOAc-CH_3OH. The HRESIMS showed a quasimolecular ion peak $[M + H]^+$ at m/z 330.1703 corresponding to the molecular formula $C_{19}H_{23}NO_4$, indicating the presence of nine degrees of unsaturation in the molecule. The ^1H, ^{13}C, DEPT135 and HSQC data of **1** showed resonances due to three olefinic methyl groups (δ_H 1.81 3H, s; 2.03 3H, s; 1.71 3H, s; δ_C 27.6, 20.6 and 16.9), a methoxy (δ_H 3.80 3H, s; δ_C 56.3), three methylenes (δ_H 2.99 2H, s; 3.35 2H, s; 4.20 2H, s; δ_C 55.0, 23.0, 43.5), an aromatic proton (δ_H 6.75 1H, s; δ_C 96.7), two trisubstituted olefins [δ_H 5.24 1H, dd, $J = 6.0$, 6.5 Hz; 6.10 1H, brs; δ_C 127.0 (CH), 129.8; 123.6 (CH), 154.9], five quaternary aromatic carbons (δ_C 119.7, 123.4, 132.1, 150.6 and 158.8) and two carbonyl groups (170.7 and 199.0). The ^1H and ^{13}C NMR data were similar to those of the known compound **6**, except for a special proton at 8.39 ppm and a methylene at 4.20 ppm which turned to be the labile proton of —NH— and the methylene connected to it. All the above data reveal the presence of an isoindolinone skeleton in **1**. The HMBC correlations as shown (H3/C1, C3a, C4, C7a; H7/C1, C3a, C6, C7a; H1′/C2′, C3′, C4, C5, C6; H6′/C5′, C8′, C10′; H9′/C2′, C3′, C4′; H10′/C6′, C7′, C8′; C6-OMe/C6) in Fig. 2 supported the analysis of the structure of **1** as shown in Fig. 1. The chemical shifts for C4 (150.6), C6 (158.8) as well as the molecular formula of **1** confirmed the substitution of a hydroxyl at C4 and a methoxyl at C6. Based on above analysis, the structure of **1** was determined as 5-[(2′E)-3′,7′-dimethyl-5′-oxo-2′,6′-dienyl]-4-hydroxy-6-methoxy-1-isoindolinone and called after erinaceolactam A.

Erinaceolactam B (**2**) was isolated as a colorless oil. The HRESIMS showed a molecular ion peak $[M]^+$ at m/z 373.1887 corresponding to

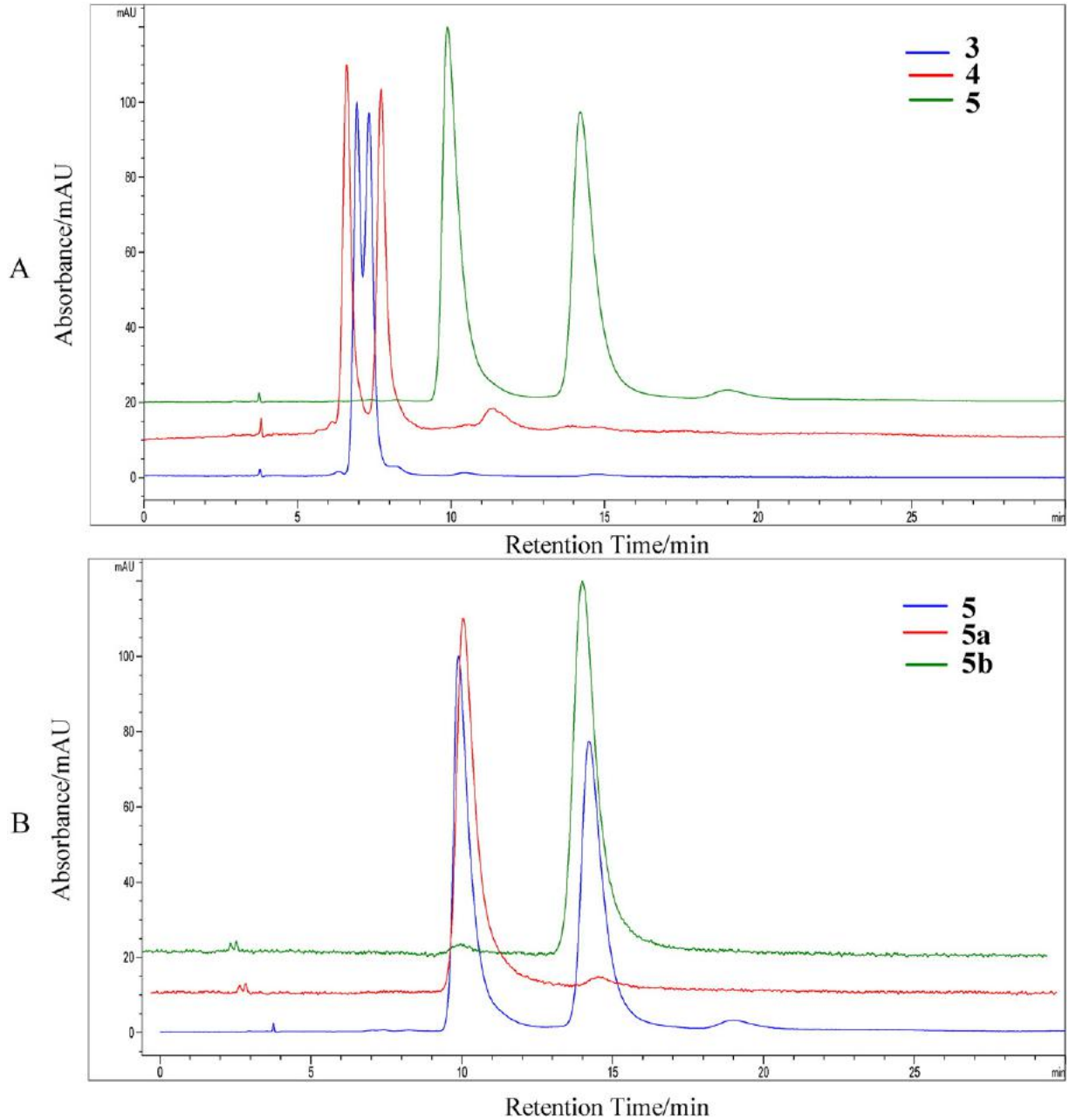
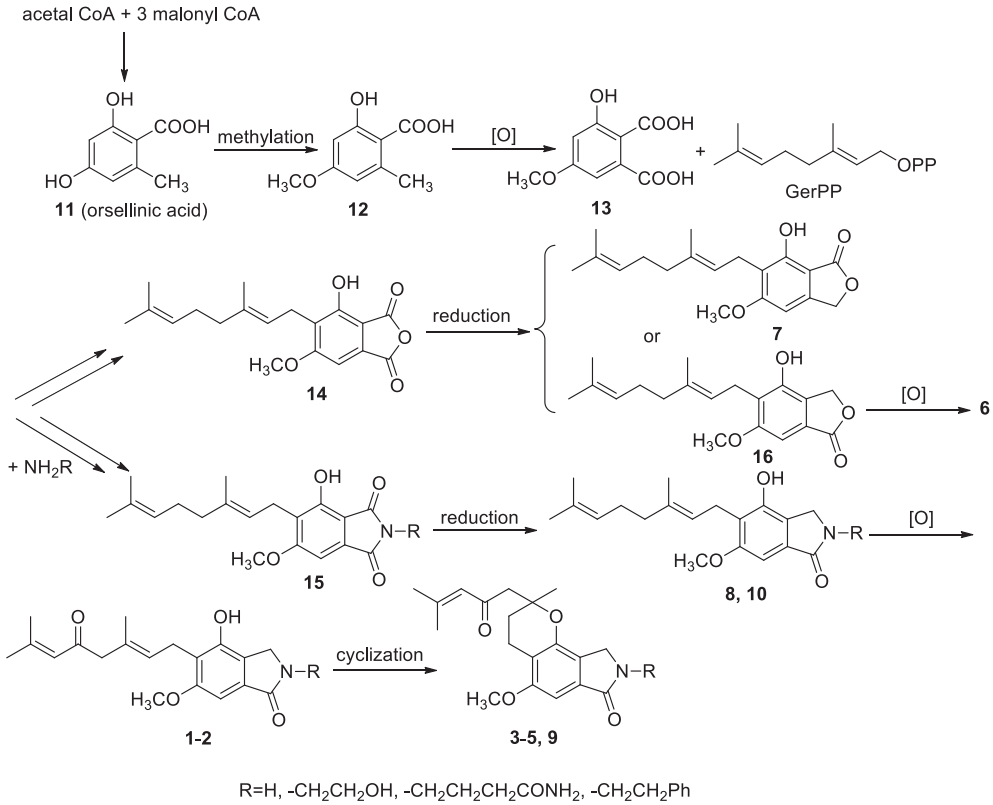


Fig. 3. Chiral HPLC separation chromatograms of compounds **3**, **4**, **5** (A) and compounds **5**, **5a**, **5b** (B) on an Acchrom S-Chiral B 5 μm 1000 A column (4.6 × 250 mmL, n-hexane- isopropyl alcohol = 8:2, 1 mL/min, detection at 254 nm).

Table 2
^{1}H and ^{13}C NMR data of compounds **6**–**10** in DMSO-d_6.

Position	**6**		**7**		**8**		**9**		**10**	
	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)
1	171.4		170.6		170.8		167.8		168.1	
2						8.37(1H, s)				
3	68.7	5.24(2H, s)	69.6	5.24(2H, s)	43.5	4.19(2H, s)	47.5	4.12(1H, d, 17.3) 4.19(1H, d, 17.3)	48.4	4.21(2H, s)
3a	123.4		148.3		132.0		122.2		134.4	
4	150.0		97.3	6.75(1H, s)	150.6		148.8		150.9	
5	122.9		164.2		120.4		113.4		120.4	
6	159.6		117.0		158.8		158.4		158.8	
7	97.9	6.89(1H, s)	154.2		96.8	6.74(1H, s)	96.2	6.76(1H, s)	96.5	6.70(1H, s)
7a	130.4		105.0		123.7		132.3		121.5	
1′	23.1	3.39(2H, d, 7.2)	21.9	3.26(2H, d, 7.0)	22.8	3.32(2H, d, 7.1)	17.4	2.61(2H, t, 6.7)	22.8	3.31(2H, d, 7.0)
2′	126.2	5.24(1H, s)	122.4	5.08(1H, t, 7.0)	122.8	5.12(1H, t, 6.9)	30.6	1.83(1H, m) 1.94(1H, m)	122.8	5.12(1H, t, 7.0)
3′	127.2	3.00(2H, s)	134.9		134.5		76.2		139.5	
4′	55.0		39.7	1.90(2H, t, 7.8)	39.6	1.90(2H, t, 8.0)	51.9	2.60(1H, d, 14.0) 2.77(1H, d, 14.0)	39.6	1.89(2H, t, 8.0)
5′	198.9		26.6	2.00(2H, m)	26.6	1.99(2H, m)	198.1		26.6	1.98(2H, m)
6′	124.4	6.09(1H, s)	124.5	5.02(1H, t, 6.8)	124.5	5.02(1H, t, 6.9)	125.5	6.10(1H, m)	124.6	5.02(1H, t, 6.9)
7′	155.0		131.1		131.1		154.9		131.7	
8′	27.6	1.81(3H, s)	25.9	1.59(3H, s)	25.9	1.58(3H, s)	27.6	1.82(3H, d, 0.9)	25.9	1.58(3H, s)
9′	20.6	2.04 (3H, s)	16.4	1.71(3H, s)	16.4	1.73(3H, s)	24.9	1.33(3H, s)	16.4	1.72(3H, s)
10′	16.9	1.71 (3H, s)	18.0	1.52(3H, s)	20.0	1.52(3H, s)	20.8	2.06(d, 0.8)	18.0	1.51(3H, s)
1″							43.6	3.74(2H, t, 14.7)	43.6	3.74(2H, t, 7.2)
2″							34.4	2.91(2H, t, 7.4)	34.5	2.89(2H, t, 7.2)
3″							139.4		139.5	
4″ or 8″							128.8	7.18–7.30(2H, m)	128.9	7.17–7.30(2H, m)
5″ or 7″							129.0	7.18–7.30(2H, m)	129.0	7.17–7.30(2H, m)
6″							126.7	7.18–7.30(1H, m)	126.7	7.17–7.30(1H, m)
4-OH		9.97(1H, s)		9.33(1H, brs)		9.42(1H, s)				9.42(1H, s)
6-OMe	56.5	3.83(3H, s)	56.7	3.85(3H, s)	56.3	3.80(3H, s)	56.2	3.83(3H, s)	56.2	3.78(3H, s)

the molecular formula $C_{21}H_{27}NO_5$, indicating the presence of nine degrees of unsaturation in the molecule. Molecular formula of **2** indicated that a fragment of $-C_2H_4O-$ was added when compared with **1**. The ^{1}H NMR showed that the signal of the proton of —NH— was lost while another labile proton appeared at 4.85 ppm. The ^{1}H combined with HSQC spectra indicated the appearance of two methylenes at 3.55 and



Scheme 1. Plausible biogenetic passway of compounds **1**–**10**.

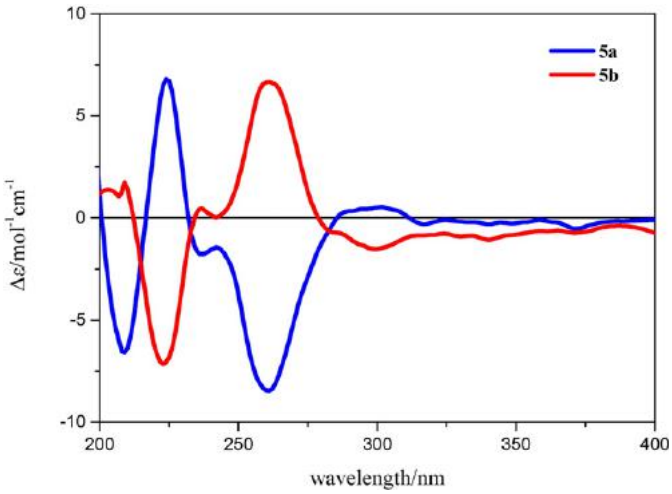


Fig. 4. Circular dichroism (CD) spectra of **5a** (c = 0.06 mg/mL, MeOH) and **5b** (c = 0.06 mg/mL, MeOH).

3.60 ppm, except these differences, the signals of ^{1}H and ^{13}C NMR were similar to those of **1**. Correlations of COSY and HMBC spectra proved that the labile proton was connected to the methylenes, while two methylenes were connected with each other. The key correlation signals were displayed in Fig. 2. In the light of spectroscopic evidence, the structure of **2** was a N-ethylol substitution of **1** and was attested as 5-[(2′E)-3′,7′-dimethyl-5′-oxo-2′,6′-dienyl]-4-hydroxy-6-methoxy-N-ethylol-1-iso indolinone. It was named as erinaceolactam B.

Erinaceolactam C (**3**) was purified as a white amorphous powder. The HRESIMS showed a quasimolecular ion peak $[M + H]^+$ at m/z 330.1704 corresponding to the molecular formula $C_{19}H_{23}NO_4$, indicating that **3** was the isomeride of **1**. Compared with **1**, the ^{1}H NMR showed that olefinic proton at 5.24 ppm disappeared, while two sets of multiplets appeared at 1.85 ppm and 1.94 ppm. The HSQC indicated that these multiplets corresponded to the same methylene. The chroman ring skeleton was elucidated by the COSY correlations between H1' and H2′ and the HMBC correlations as shown in Fig. 2 (H7/C7a, C6, C5, C1; H1′/C4, C5, C6, C2′; H2′/C1′, C3′, C5; H9′/C2′, C3′, C4′; C6-*OMe*/C6). The side chain moiety attached to C3′ was elucidated by the HMBC correlations (H4′/C2′, C3′, C5′, C9′; H6′/C8′, C10′, C5′; H8′/C7′, C10′, C6′; H10′/C8′, C7′, C6′). The γ-lactam was determined by the HMBC correlations (H3/C4, C3a, C1, C7a; H7/C1, C5, C6, C7a; H2/C1, C3, C3a, C7a). In the light of spectroscopic evidence, **3** was the cyclization product of **1** as displayed and they were a pair of isomerides. The key correlation signals were displayed in Fig. 2. The structure of **3** was certified as 3′-(7′-methyl-5′-oxo-6′-enyl)-3′-methyl-6-methoxy-1′,2′-dihydro-3′*H*-pyrrole[3.4-*h*]chromen-1-(3*H*)-one and designated as erinaceolactam C. It was suggested to be racemic by the specific rotation data.

Erinaceolactam D (**4**) was obtained as a yellowish amorphous powder. The HRESIMS showed a quasimolecular ion peak $[M + H]^+$ at m/z 374.1965 corresponding to the molecular formula $C_{21}H_{27}NO_5$, indicating that **4** was the isomeride of **2**. Compared with **3**, the molecular formula of **4** revealing that a fragment of —C_2H_4O— was added. The ^{1}H NMR showed that the signal of the proton of —NH— was lost while another labile proton appeared at 4.83 ppm. The ^{1}H combined with HSQC NMR spectrum indicated the appearance of two methylenes at 3.59 and 3.54 ppm. Except these differences, the signals of ^{1}H NMR and ^{13}C NMR were similar to those of **3**. Correlations of COSY and HMBC spectra proved that the labile proton was connected to the methylenes, while two methylenes were connected with each other. In the light of spectroscopic evidence, **4** was a N-ethylol substitution of **3**. The key correlation signals were displayed in Fig. 3. On the basis of above analysis, the structure of **4** was determined as 3′-(7′-methyl-5′-oxo-6′-enyl)-3′-methyl-6-methoxy-*N*-ethylol-1′,2′-dihydro-3′*H*-pyrrole[3.4-*h*]chromen-1-(3*H*)-one and named after erinaceolactam D. It was assigned to be racemic by the specific rotation data.

Erinaceolactam E (**5**) was gained as a yellow amorphous powder. The HRESIMS showed a quasimolecular ion peak $[M + H]^+$ at m/z 415.2245 corresponding to the molecular formula $C_{23}H_{30}N_2O_5$, indicating the presence of ten degrees of unsaturation in the molecule. Compared with **3**, the molecular formula of **5** revealed that a fragment of —C_4H_7NO— was added. The ^{1}H NMR showed that the signal of the proton of —NH— was lost while another two labile protons appeared at 6.74 ppm and 7.28 ppm. It can be elucidated by the COSY correlations that these two labile protons were near to each other. Besides, three more methylenes appeared. They turns to form the side chain of n-butyramido by the COSY correlations (H1″/H2″; H2″/H3″) and the HMBC correlations (H1″/C1, C3, C2″, C3″; H2″/C1″, C3″; H3″/C1″, C2″, C4″). The chroman skeleton and γ-lactam moiety was elucidated as identical with that of **3** in the similar manner. In the light of spectroscopic evidence, the structure of **5** was a N-n-butyramido substitution of **3**. The key correlation signals were displayed in Fig. 2. Based on above analysis, the structure of **5** was testified as 3′-(7′-methyl-5′-oxo-6′-enyl)-3′-methyl-6-methoxy-N-n-butyramido-1′,2′-dihydro-3′*H*-pyrrole[3.4-*h*]chromen-1-(3*H*)-one and called after erinaceolactam E. Erinaceolactam E was determined to be racemic by the specific rotation data.

Five known compounds, hericenone A (**6**) [17], hericenone J (**7**) [18], *N*-De phenylethyl isohericerin (**8**) [4], erinacerin A (**9**) [19], and hericerin (**10**) [20], were identified by comparison of their NMR data (Table 2) with the literature data. A plausible biogenetic pathway to isoindolinones and their analogues are proposed as shown in Scheme 1. Biogenetically, orsellinic acid (**11**) derived from acetyl malonyl pathway is the preliminary substrate [14]. Orsellinic acid is first methylated and oxidated into **13**. Compound **13** reacts with geranyl pyrophosphate (GerPP) or with geranyl pyrophosphate (GerPP) and amine to give phthalic anhydride (**14**) or phthalimide (**15**). Then, compound **14** could derive to compound **7** or **16** by reduction at different position. Meanwhile, compound **15** could derive to compounds **1**–**2** by reduction, and farther to **3**–**5**, **9** by cyclization.

The cyclization between the olefinic carbon at C-3′ and hydroxyl group at C-4 generates the dihydropyran ring in the molecular

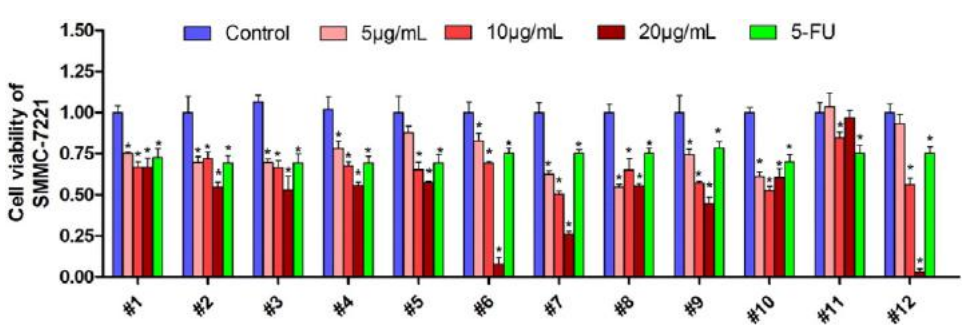


Fig. 5. Effects of compounds **1**–**10**, petroleum ether extract (**11**), EtOAc extract (**12**) and positive control 5-FU (5 μg/mL) on cell viability of SMMC-7221. Data are mean ± S.E.M. n = 3. $^*P < 0.05$ vs. Control.

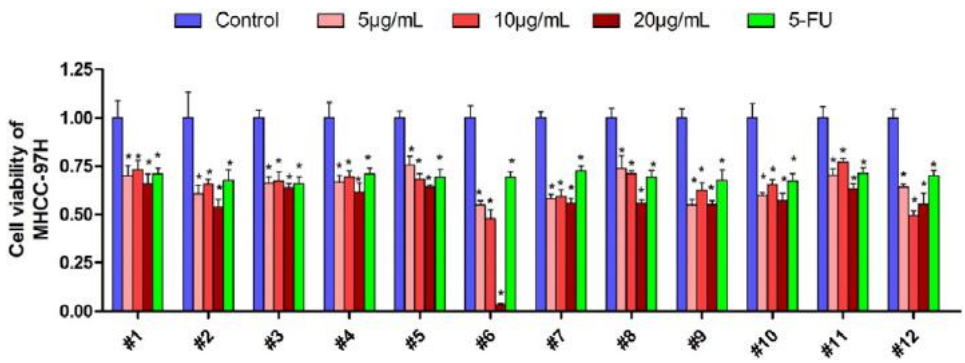


Fig. 6. Effects of compounds **1**–**10**, petroleum ether extract (**11**), EtOAc extract (**12**) and positive control 5-FU (5 μg/mL) on cell viability of MHCC-9. Data are mean ± S.E.M. $n = 3$. $^{*}P < 0.05$ vs. Control.

structures of compounds **3**–**5**, **9**. It also leads to the presence of an asymmetric carbon at C-3′. The specific rotations of compounds **3**–**5**, **9** were almost zero, suggesting that they were racemic. Chiral separations were performed by chiral HPLC, in which all of three new compounds **3**–**5** were separated into two peaks with equal integral proportions as shown in Fig. 3. Compound **5** was chosen as the representative one to be chiral Pre-HPLC separated to give **5a** [1.2 mg, $[\alpha]_D^{25} = +21.6$ ($c = 0.2$, MeOH)] and **5b** [1.2 mg, $[\alpha]_D^{25} = -21.1$ ($c = 0.2$, MeOH)] finally. The circular dichroism (CD) spectra of **5a** and **5b** displayed in Fig. 4 exhibited mirror image CD curves, which proved that **5a** and **5b** were enantiomers. The chromatograms of chiral separations and CD spectra also confirmed the racemism of compounds **3**–**5** and the cyclization without stereoselectivity.

3.2. Biological activity

The anticancer activity of the extracts from *H. erinaceus* has been investigated previously [15]. In order to screen out bioactive metabolites from the extracts of *H. erinaceus*, the cytotoxicity activity of samples **1**–**12** against human liver cancer cell line SMMC-7221 and MHCC-97H were evaluated. As shown in Fig. 5, all of the samples **1**–**12** exhibited significant cell growth inhibition against SMMC-7221 comparable to or more potent than 5-FU. Compound **1**, **3**, **4**, **5**, **6**, **7**, **9** and EtOAc extract (**12**) inhibited the growth of SMMC-7221 in a dose-dependent manner. In MHCC-97H, compound **5**, **6** and **8** dose-dependently inhibited the cell growth as shown in Fig. 6. Among these compounds **1**–**10**, compound **6** (20 μg/mL) showed the most potent activity against the growth of SMMC-7221 and MHCC-97H, which might be an efficient lead compound for cancer therapy. It was noteworthy that EtOAc extract (**12**) and petroleum ether extract (**11**) exhibited comparable activity, these results suggested that other anticancer activity substances may exist in *H. erinaceus* in addition to compound **1**–**10** and which would be next work focus.

In summary, five new isoindolinone alkaloids, erinaceolactams A–E, were isolated from 70% ethanol extract of fruiting bodies of *H. erinaceus*. It was interesting to compare erinaceolactam A with C and erinaceolactam B with D that the latter was the cyclization product of the former. They are two pairs of isomers, which are firstly discovered simultaneously in *H. erinaceus*. There might be some enzymes to mediate the cyclization process which deserved our further study. It was also demonstrated that isoindolinones were the main active small molecules in *H. erinaceus*. The test of antitumor for all isolates was conducted by MTT methods, which proved that some of the compounds would inhibit the growth of human liver cancer SMMC-7221 and MHCC-97H cells. The results suggested that isoindolinone alkaloids contributed to the antitumor effects of *H. erinaceus* extract.

Acknowledgments

We wish to express our gratitude to Prof. Shaogang Liu for NMR measurements and Dr. Hongping Long for MS measurements. The work was supported by grants from National Natural Science Foundation of China (No. 31370370), Key Researching and Transforming Project of Science and Technology of strategic emerging industry of Hunan Province (No. 2015GK1053), Key R&D Project of applied technology of Haikou City (No. 2015039), and the Science and Technology Planning Project of Hunan Province, China (No. 2014SK2002, 2013SK5077).

Appendix A. Supplementary data

The 1D (^{1}H, ^{13}C and DEPT135 NMR), 2D NMR (^{1}H—^{1}H COSY, HSQC and HMBC), HRESIMS and MS/MS spectra of new compounds are available in the Supplementary data.

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